Golden Eras

Time limit per test: 2 seconds
Memory limit per test: 1 gb

On the evening of 6 October 2026, an old storyteller sat by the fire and opened a scrapbook he had kept for twenty years. "This is a fairy tale," he told his granddaughter, "about a little boy with a magic left foot."

The tale began in the summer of 2006, when the boy first stepped onto the biggest stage in the world, and it ended in the summer of 2026. In between, the whole world gathered every four years: 2006, 2010, 2014, 2018, 2022 and 2026.

The scrapbook has **N** pages, numbered from **1** to **N**. On page **i** the storyteller wrote a year a_i. Some pages were glued in a hurry, so the years are not always in order, some of them repeat, and some have nothing to do with the tale at all.

The granddaughter loves the **golden eras** of the tale. A golden era is a stretch of consecutive pages **x**, **x** + 1, ..., **y** with **x** < **y** (so it has at least two pages), in which every page shows a year exactly 4 greater than the page right before it:

$$a_{k+1} = a_k + 4 \quad \text{for every } \mathbf{k} \text{ with } \mathbf{x} <= \mathbf{k} < \mathbf{y}$$

For example, three consecutive pages showing 2006, 2010, 2014 form a golden era, and so do their first two pages and their last two pages. Two consecutive pages showing 2010, 2006 do not (the years go backwards), and neither do 2014, 2022 (eight years apart) or 2018, 2018.

She asks **Q** questions. In each question she points at the pages from **L** to **R** and asks how many golden eras lie completely inside them, that is, how many pairs (**x**, **y**) with **L** <= **x** < **y** <= **R** are such that the pages **x**, **x** + 1, ..., **y** form a golden era.

Help the storyteller answer every question before the fire goes out.

Input:

The first line contains a single integer **t** (1 <= **t** <= 10^4) representing the number of test cases.

Each test case consists of:

1) A line containing two integers **n** and **q** (1 <= **n**, **q** <= 2 x 10^5) representing the number of pages and the number of questions.

2) A line containing **n** integers a_1, a_2, ..., a_n (1 <= a_i <= 10^9) representing the years written on the pages.

3) **q** lines, each containing two integers **L** and **R** (1 <= **L** <= **R** <= **n**) describing one question.

It is guaranteed that the sum of n over all test cases does not exceed 2 x 10^5, and the sum of q over all test cases does not exceed 2 x 10^5.

Output:

For each question, print a single integer on its own line: the number of golden eras that lie completely inside the pages from **L** to **R**.

Print the answers in the order the questions are given, test case after test case.

Example:

Input
2 6 3 2006 2010 2014 2018 2022 2026 1 6 2 4 3 3 8 3 2014 2010 2014 2018 2018 2022 2030 2034 1 8 3 6 1 2
Output
15 3 0 5 2 0

Explanation:

Test case 1

Question [1, 6]: the six pages show 2006, 2010, 2014, 2018, 2022, 2026, so any two pages **x** < **y** enclose a golden era. There are 6 x 5 / 2 = 15 such pairs.

Question [2, 4]: the pages show 2010, 2014, 2018. The golden eras are pages 2 to 3, pages 3 to 4 and pages 2 to 4, so the answer is 3.

Question [3, 3]: a single page is never a golden era, so the answer is 0.

Test case 2

Question [1, 8]: the golden eras are pages 2 to 3, 3 to 4 and 2 to 4 (2010, 2014, 2018), pages 5 to 6 (2018, 2022) and pages 7 to 8 (2030, 2034), so the answer is 5. Pages 1 to 2 go backwards, pages 4 to 5 show the same year, and pages 6 to 7 are eight years apart.

Question [3, 6]: the golden eras completely inside are pages 3 to 4 and pages 5 to 6, so the answer is 2. Pages 2 to 4 form a golden era of the scrapbook, but page 2 is outside this question.

Question [1, 2]: the pages show 2014, 2010, which go backwards, so the answer is 0.

Notes:

The years on the pages do not have to be World Cup years; only the difference between consecutive pages matters.

Questions do not change the scrapbook; each question is answered independently.